

COMPARATIVE ANALYSIS OF RAG TECHNIQUES AND LLM MODELS FOR PERSONALIZED NEWS SUMMARIZATION

Abstract

The exponential growth of digital news content has created challenges for users in consuming and understanding large volumes of information efficiently. Personalized news summarization aims to provide concise and relevant summaries based on user interests and queries. This project presents a comparative analysis of Retrieval-Augmented Generation (RAG) techniques and Large Language Models (LLMs) for personalized news summarization. Three retrieval approaches, namely Dense Retrieval, Hybrid Retrieval, and Fusion Retrieval, are evaluated using GPT, Llama, and Qwen language models. The system retrieves relevant news articles, generates summaries, and evaluates performance using ROUGE and BERTScore metrics. Experimental results demonstrate the effectiveness of combining advanced retrieval mechanisms with modern language models to improve summary quality and contextual relevance.

1. Introduction

With the increasing availability of online news articles, users often struggle to process large amounts of information. Automatic text summarization helps reduce reading time while retaining essential information. However, traditional summarization methods often lack personalization and contextual understanding.

Retrieval-Augmented Generation (RAG) addresses these limitations by retrieving relevant documents before generating summaries. By providing external knowledge to language models, RAG improves factual accuracy and contextual relevance. This project investigates the impact of different retrieval strategies and language models on personalized news summarization performance.

2. Dataset

The project utilizes the CNN/DailyMail dataset, a widely used benchmark for text summarization research.

Dataset Characteristics

- Large collection of news articles
- Human-written reference summaries
- Multiple news categories
- Suitable for summarization and retrieval evaluation

Preprocessing Steps

- Removal of unwanted characters
- Text normalization
- Document chunking

- Tokenization
- Embedding generation

3. Methodology

The proposed methodology consists of the following phases:

Phase 1: Data Collection

News articles are loaded from the CNN/DailyMail dataset.

Phase 2: Data Preprocessing

Articles are cleaned, normalized, and divided into smaller chunks for efficient retrieval.

Phase 3: Embedding Generation

Sentence Transformer models generate dense vector embeddings for all document chunks.

Phase 4: Retrieval

Relevant documents are retrieved using:

- Dense Retrieval
- Hybrid Retrieval
- Fusion Retrieval

Phase 5: Summary Generation

Retrieved documents are passed to selected LLMs (GPT, Llama, or Qwen) to generate summaries.

Phase 6: Evaluation

Generated summaries are evaluated using standard metrics.

4. RAG Techniques

4.1 Dense Retrieval

Dense Retrieval uses vector embeddings and semantic similarity search. It captures contextual meaning and retrieves semantically related documents even when exact keywords are absent.

Advantages:

- Strong semantic understanding
- Effective contextual retrieval

Limitations:

- Computationally expensive
- Requires embedding generation

4.2 Hybrid Retrieval

Hybrid Retrieval combines dense semantic search with keyword-based retrieval methods.

Advantages:

- Improved retrieval accuracy
- Better handling of rare keywords
- Balanced retrieval performance

4.3 Fusion Retrieval

Fusion Retrieval combines outputs from multiple retrieval systems using score aggregation techniques.

Advantages:

- Increased diversity of retrieved documents
- Better overall relevance
- Improved robustness

5. Large Language Models

GPT

GPT is a powerful generative language model capable of producing coherent and context-aware summaries. It demonstrates strong performance in text generation tasks.

Llama

Llama is an open-source language model optimized for efficient inference and high-quality language understanding.

Qwen

Qwen is a multilingual language model capable of generating fluent and informative summaries while supporting multiple languages.

6. Evaluation Metrics

ROUGE-1

Measures unigram overlap between generated and reference summaries.

ROUGE-2

Measures bigram overlap.

ROUGE-L

Measures longest common subsequence similarity.

BERTScore

Evaluates semantic similarity using contextual embeddings.

Retrieval Relevance

Measures the quality of retrieved documents.

Overall Ranking Score

Combines multiple evaluation metrics for final comparison.

7. Results and Analysis

Experiments were conducted using all combinations of retrieval techniques and language models.

Observations

- Dense Retrieval provided strong semantic relevance.
- Hybrid Retrieval achieved balanced performance.
- Fusion Retrieval produced more diverse contextual information.
- GPT generated highly coherent summaries.
- Llama provided efficient and consistent results.
- Qwen demonstrated competitive multilingual capabilities.

Performance was visualized using ranking charts and heatmaps to compare all model combinations.

8. Best Combination

Based on evaluation metrics and ranking scores, the best-performing combination achieved superior retrieval quality and summary generation performance.

Key Factors:

- High ROUGE scores
- Strong BERTScore values
- Better contextual relevance
- Consistent retrieval accuracy

The selected combination is recommended for deployment in real-world personalized news summarization systems.

9. Conclusion

This project demonstrates the importance of integrating Retrieval-Augmented Generation techniques with Large Language Models for personalized news summarization. The comparative study reveals that retrieval quality significantly influences summary performance, while language models contribute to fluency and coherence. The findings provide valuable insights for designing intelligent news recommendation and summarization systems.

10. Future Work

Future enhancements include:

- Real-time news streaming integration
- Personalized user profiling
- Multilingual summarization
- Adaptive retrieval mechanisms
- Advanced reranking techniques
- Cloud deployment using Streamlit and Docker
- Mobile application integration

References

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